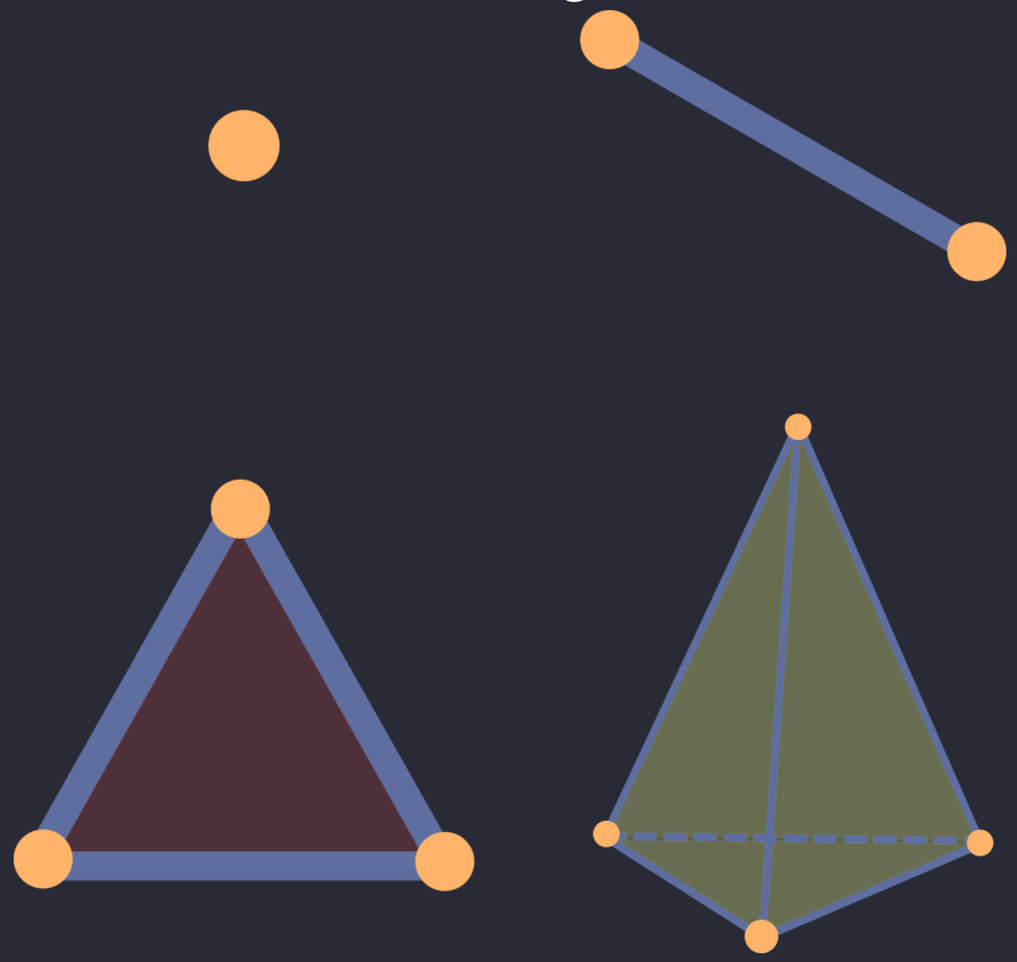
DISCRETE MORSE THEORY

SIMPLICIAL COMPLEX

SIMPLEX

A simplex is just an n-dimensional “triangle”



formally

$$\Delta^n = \left\{ x \in \mathbb{R}^k : x_0 + \cdots + x_{k-1} = 1, x_i \geq 0 \text{ for } i = 0, \ldots, k-1 \right\}$$

We can also describe an n simplex combinatorially as a list of the vertices of the simplex

$$\sigma = [e_0,\ e_1,\ \cdots, e_n]$$

FACE OF A SIMPLEX

We define a face τ, of a simplex σ, as a proper subset of σ that is

$$\tau < \sigma$$

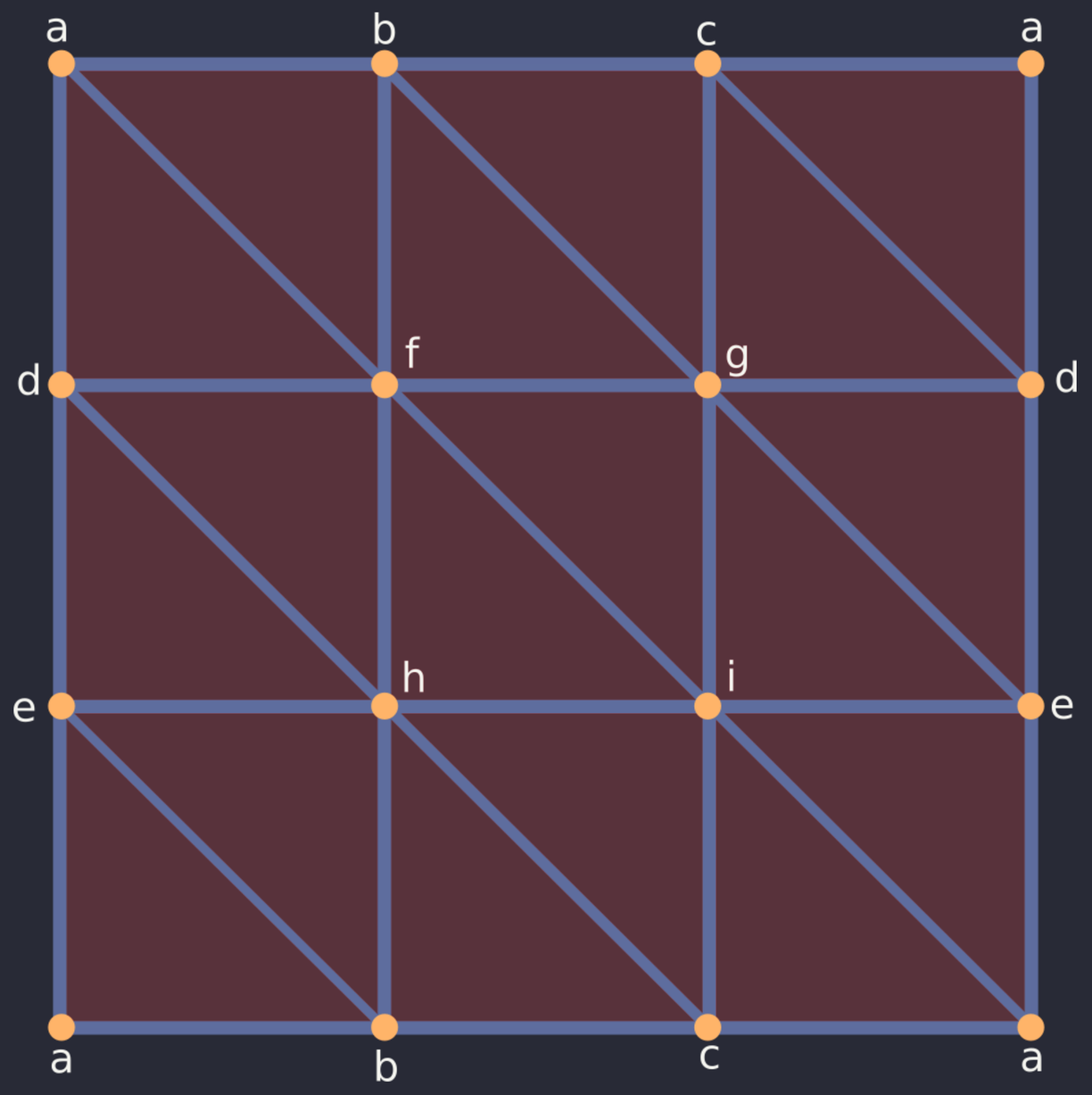
when

$$[a, b] \subset [a,\ b,\ c]$$

SIMPLICIAL COMPLEX

A simplicial complex is a space K built from glued together simplices and must satisfy

1. If τ is the face of a simplex $\sigma \in K$ then $\tau \in K$
2. If $\sigma, \tau \in K$ then $\sigma \cap \tau$ is a face of σ and τ.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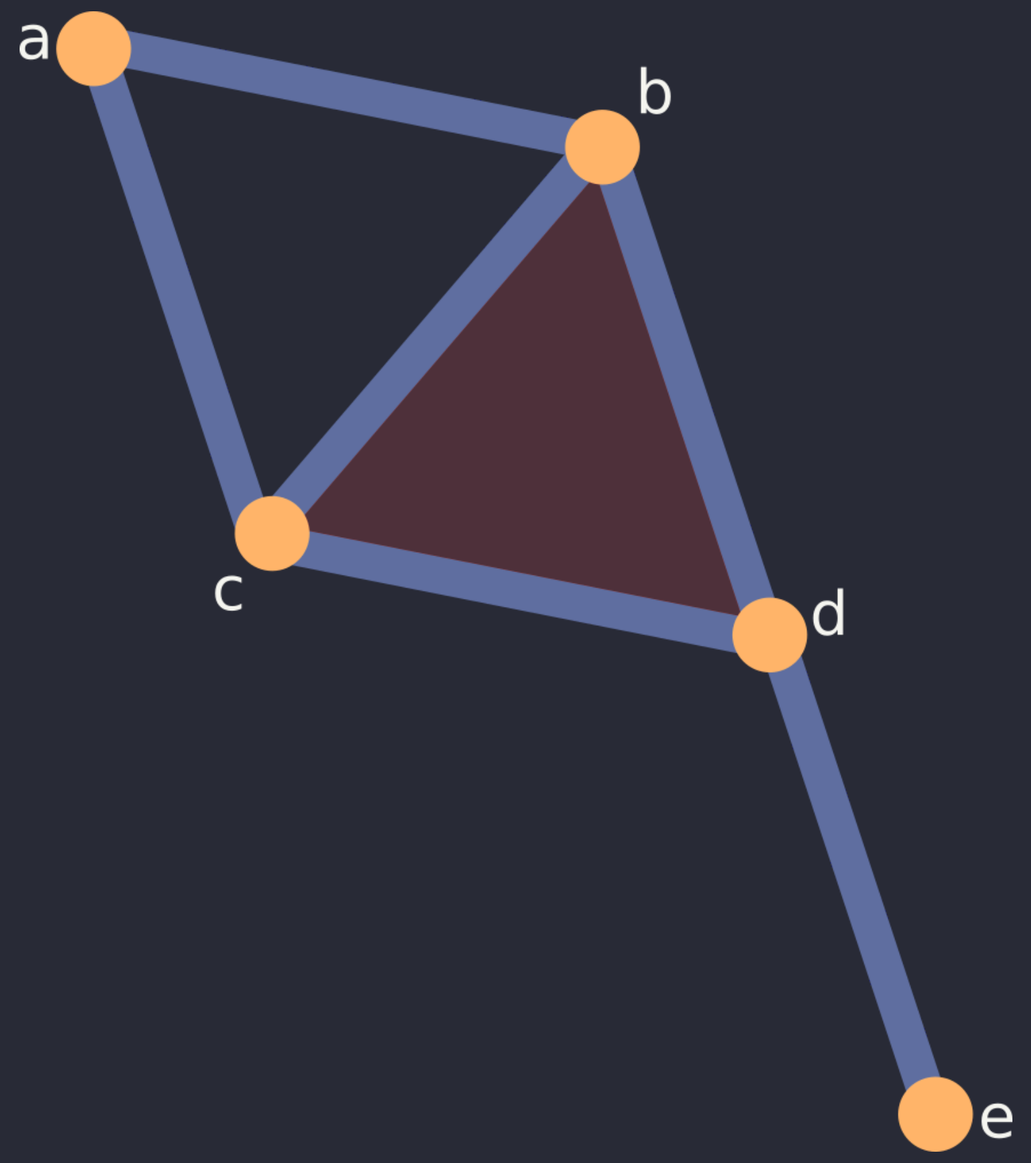
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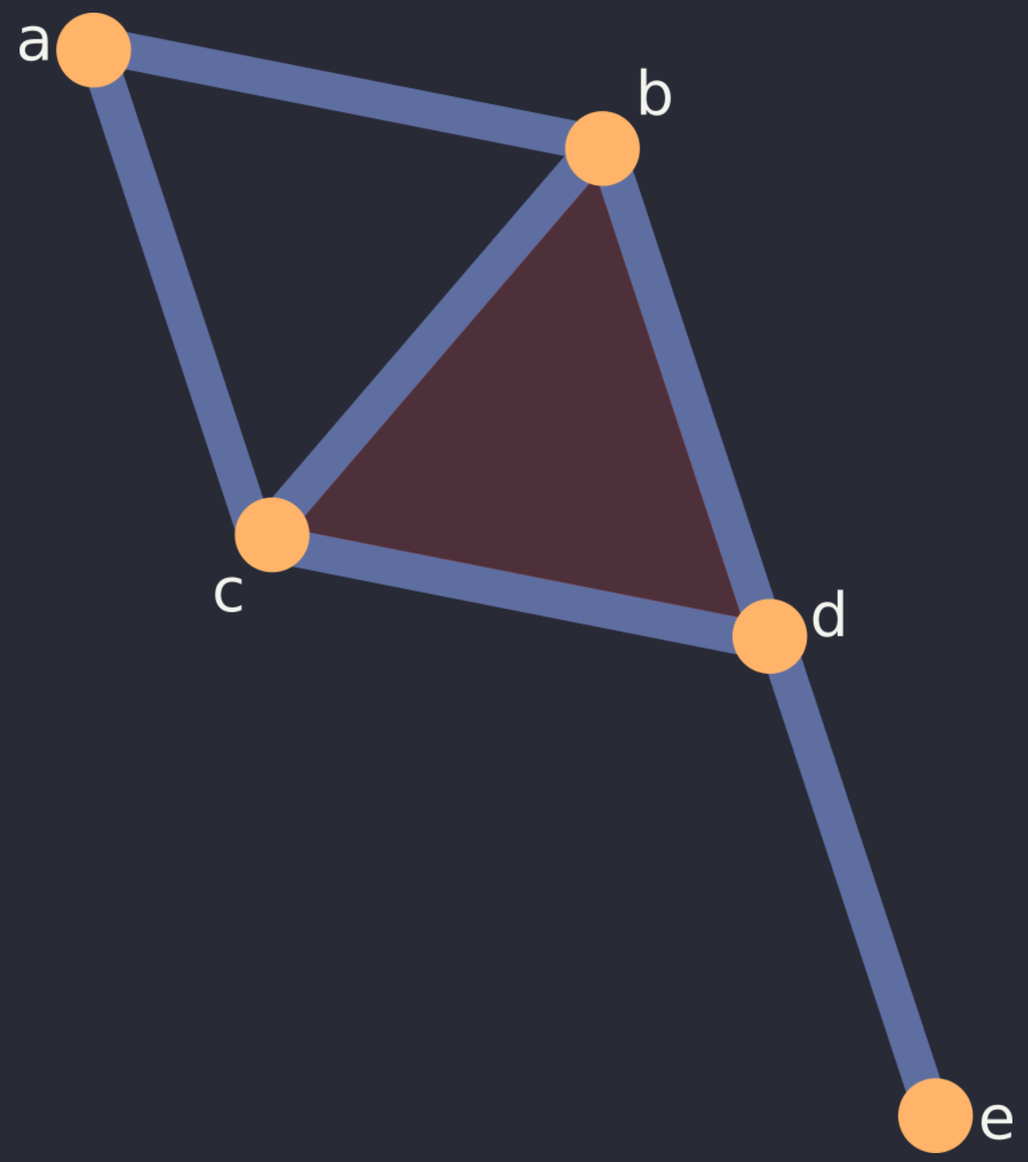
It's convenient to have a combinatorial description for K. This description is called an "abstract simplicial complex" and is just the collection of combinatorial descriptions of simplices of K

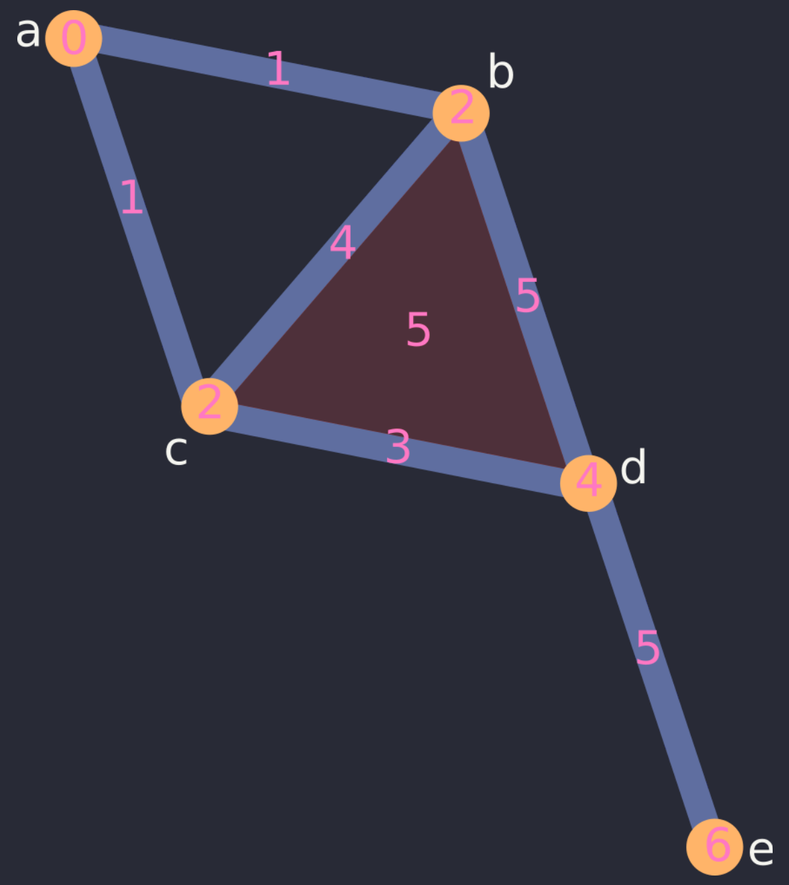


$$\{[b,c,d],[b,c],[b,d],[c,d],[a,c],[a,b],[d,e],[a],[b],[c],[d],[e]\}$$

DEFINITION OF A “DISCRETE FUNCTION”

A discrete function f on a simplicial complex K assigns a number to each simplex

$$
\begin{aligned}
f([b,c,d]) &= 5 \\
f([b,c]) &= 4 \\
f([b,d]) &= 5 \\
f([c,d]) &= 3 \\
f([a,c]) &= 1 \\
f([a,b]) &= 1 \\
f([d,e]) &= 5 \\
f([a]) &= 0 \\
f([b]) &= 2 \\
f([c]) &= 2 \\
f([d]) &= 4 \\
f([e]) &= 6
\end{aligned}
$$



DEFINITION OF A "DISCRETE MORSE FUNCTION"

A discrete function

$$f : K \to \mathbb{R}$$

is called a discrete Morse function if for every n-simplex $\alpha^n \in K$ the following are true

1. $\#\left\{\alpha^n < \beta^{n+1} \;\middle|\; f\left(\beta^{n+1}\right) \leq f\left(\alpha^n\right)\right\} \leq 1$
2. $\#\left\{\alpha^n > \gamma^{n-1} \;\middle|\; f\left(\gamma^{n-1}\right) \geq f\left(\alpha^n\right)\right\} \leq 1$

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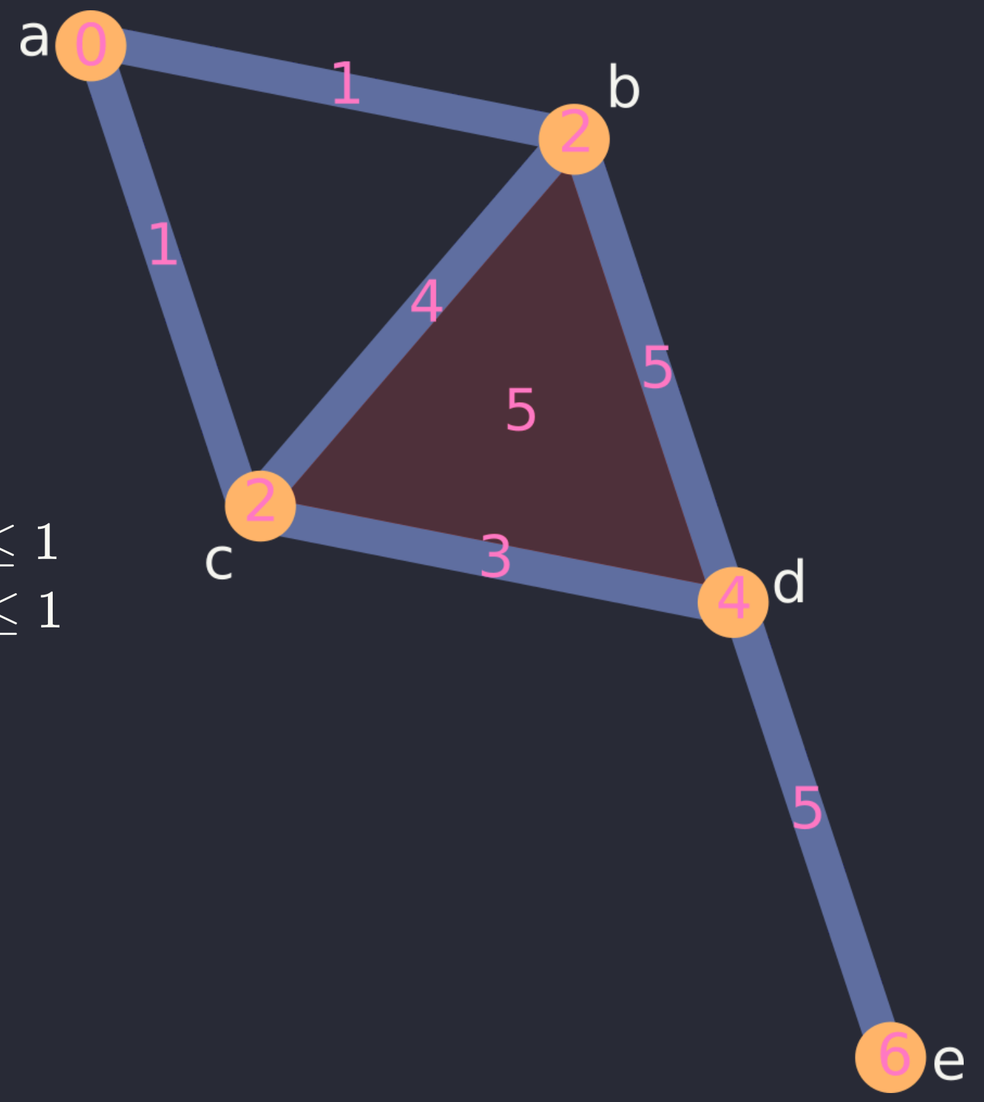

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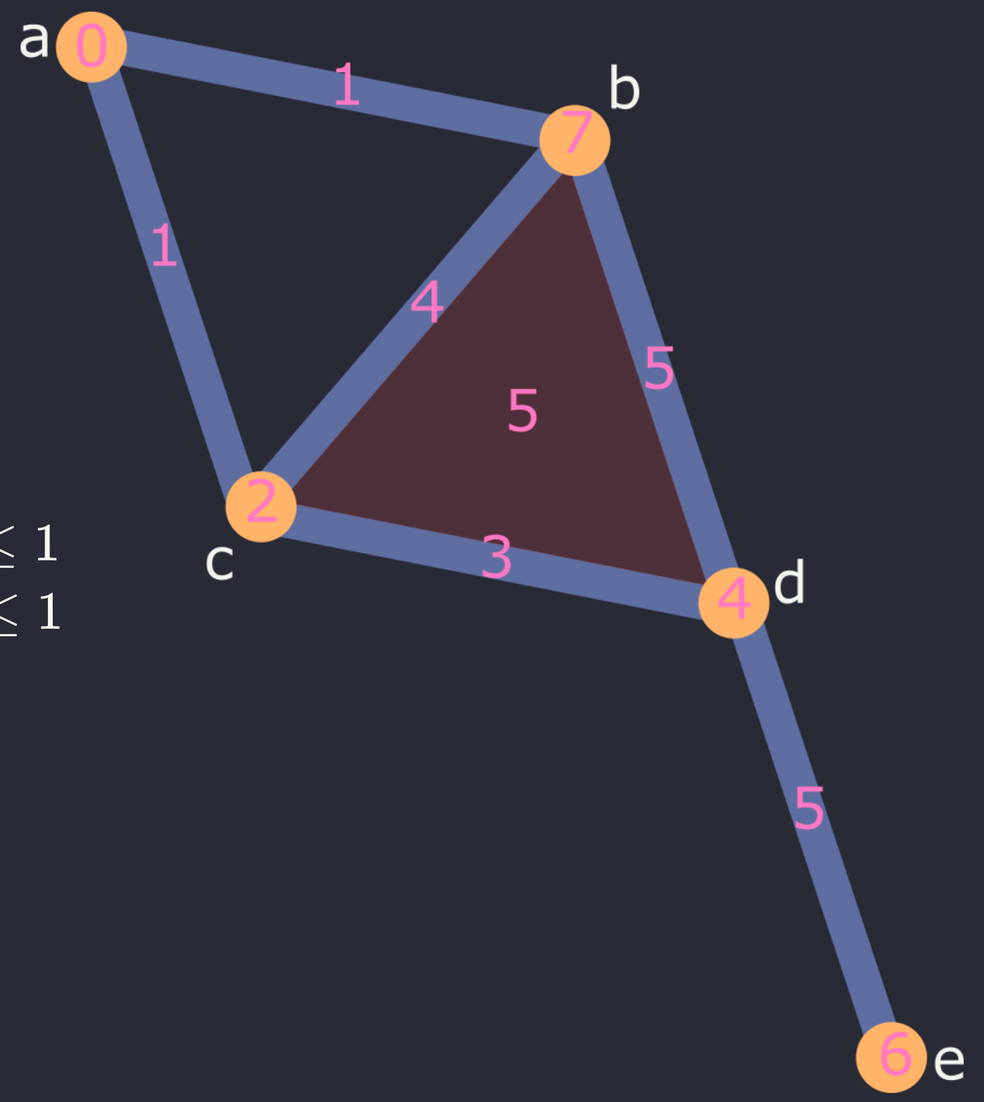
2. $\#\left\{\alpha^n > \gamma^{n-1} \;\middle|\; f\left(\gamma^{n-1}\right) \geq f\left(\alpha^n\right)\right\} \leq 1$



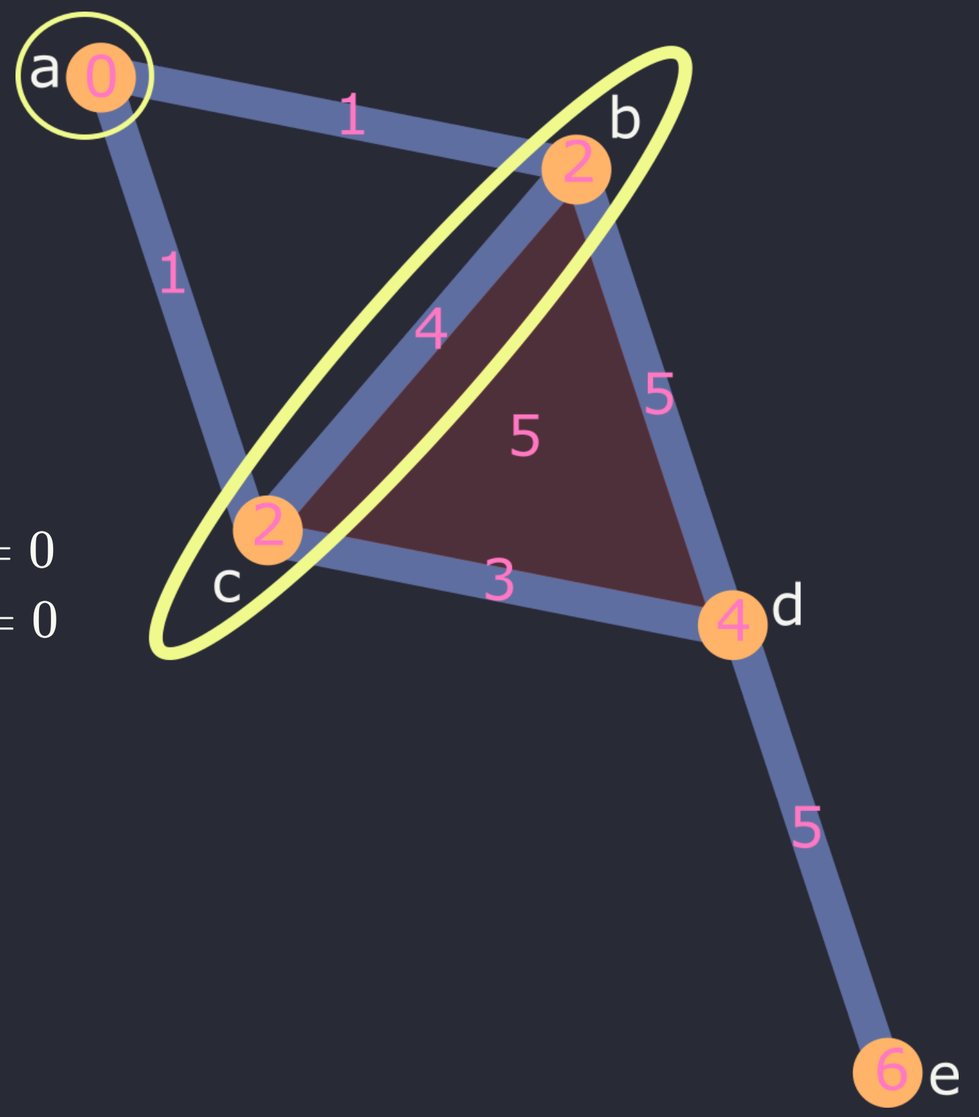
DEFINITION OF A CRITICAL SIMPLEX

An $n-$simplex α^n, of a complex K, is critical under a discrete Morse function f if the following are true

$$
\begin{aligned}
&1.\ \#\left\{\alpha^n < \beta^{n+1} \;\middle|\; f\left(\beta^{n+1}\right) \leq f\left(\alpha^n\right)\right\} = 0 \\
&2.\ \#\left\{\alpha^n > \gamma^{n-1} \;\middle|\; f\left(\gamma^{n-1}\right) \geq f\left(\alpha^n\right)\right\} = 0
\end{aligned}
$$

We should note that for all the $\alpha \in K$ at least one of $1.$ and $2.$ must be true.

1. $\#\left\{\alpha^n < \beta^{n+1} \;\middle|\; f\left(\beta^{n+1}\right) \leq f\left(\alpha^n\right)\right\} = 0$
2. $\#\left\{\alpha^n > \gamma^{n-1} \;\middle|\; f\left(\gamma^{n-1}\right) \geq f\left(\alpha^n\right)\right\} = 0$



DISCRETE GRADIENT VECTOR FIELD

A discrete gradient vector field on a simplicial complex K is a collection of pairs $V = \left\{\tau^n < \sigma^{n+1}\right\}$ of simplices in K such that each simplex is in at most one pair of V.

A discrete Morse function induces a gradient vector field on K.

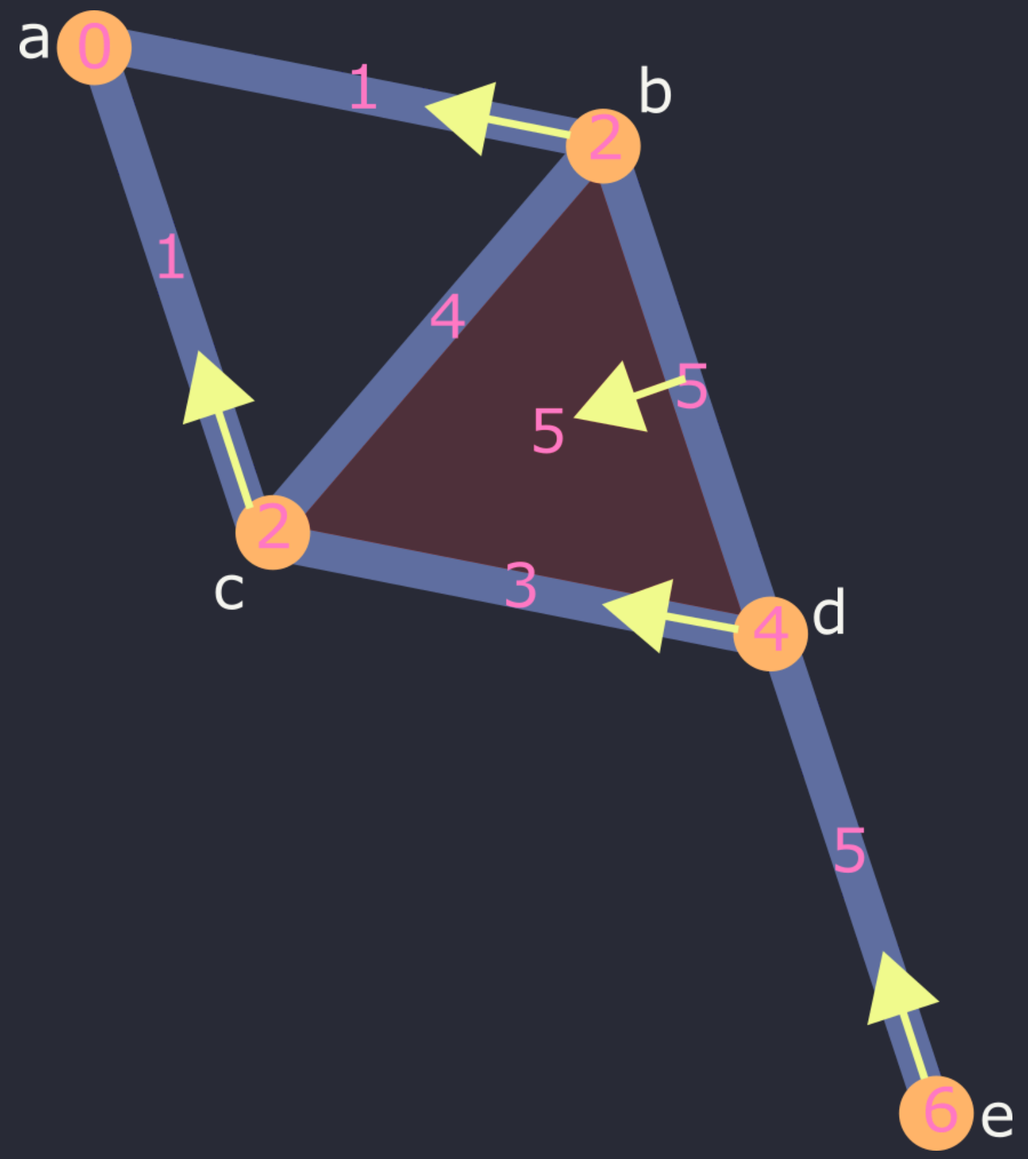
If α is a non-critical simplex, and

$$\alpha < \beta$$

with

$$f(\beta) < f(\alpha)$$

We draw an arrow from $\alpha \to \beta$.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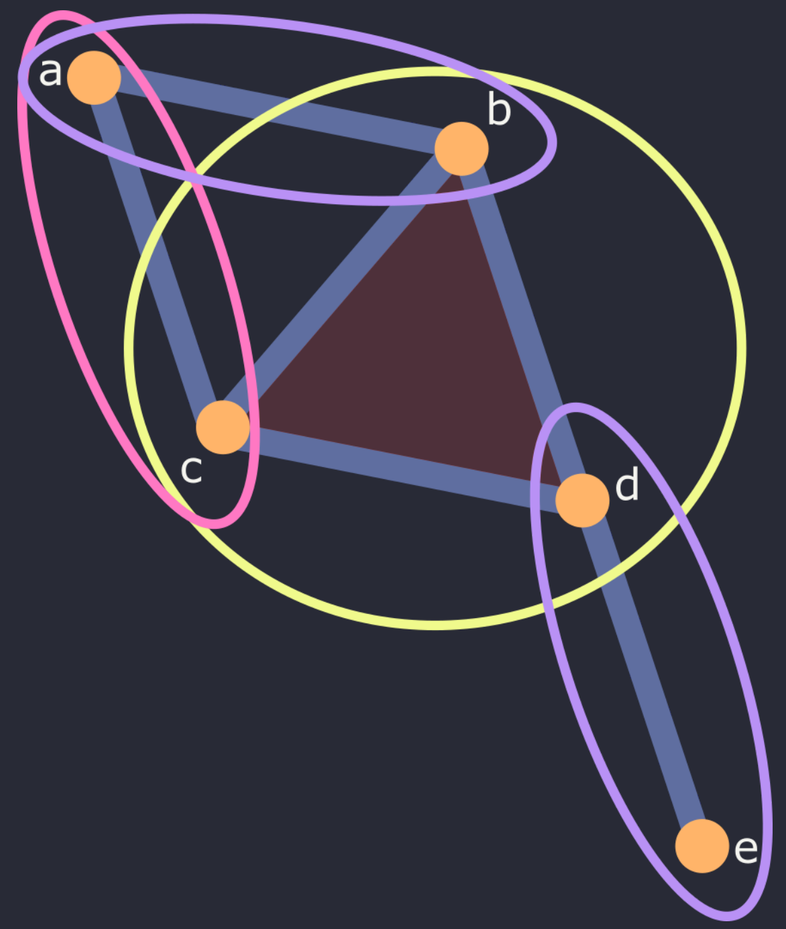
SIMPLICIAL COLLAPSE

The simplicial equivalent of a deformation retraction is called a simplicial collapse.

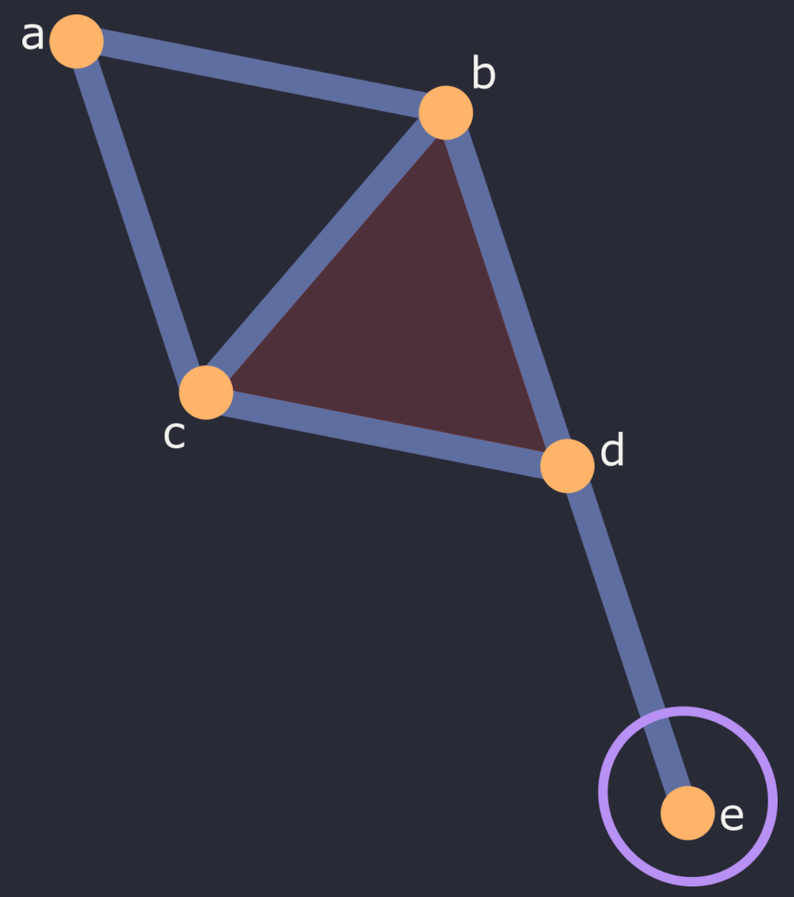
MAXIMAL FACE

A face $\sigma \in K$ is called maximal if there is no $\beta \in K$ so that $\sigma < \beta$.



FREE FACE

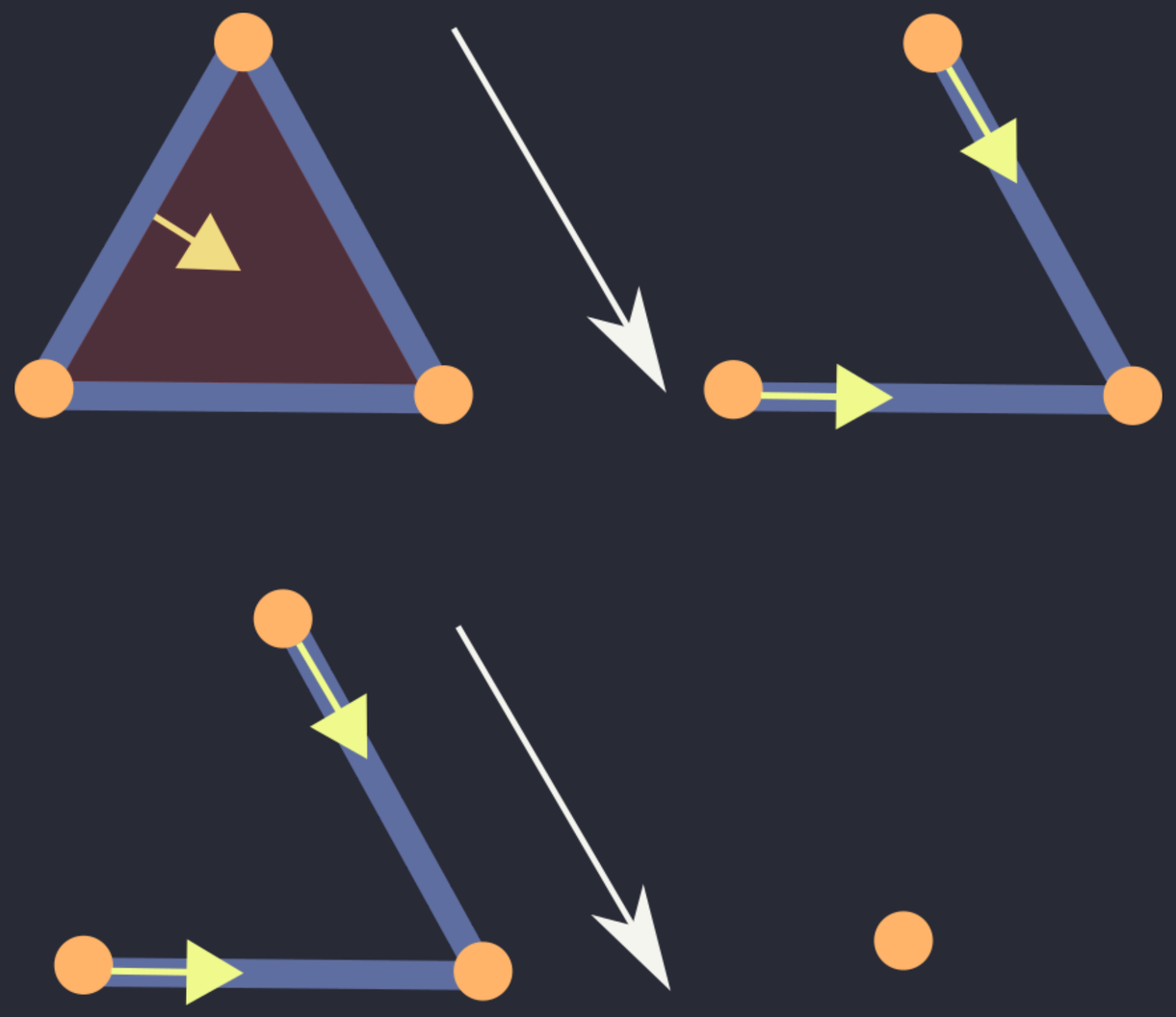
A simplex τ of a simplicial complex K is called a free face if τ is a face of exactly one maximal face.



COLLAPSE

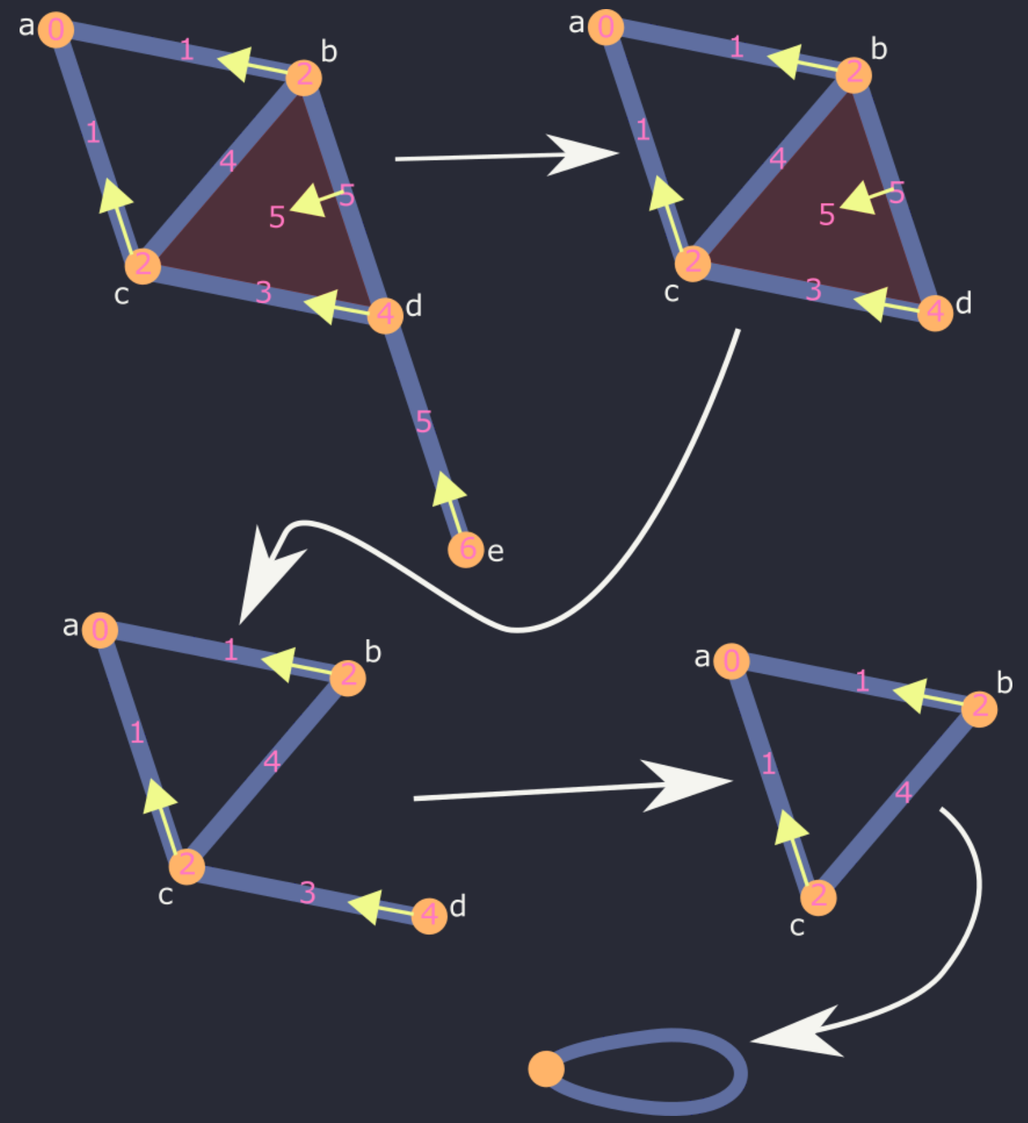
If τ is a free face of σ we can collapse $K \searrow K'$ by the following set operation

$$K' = K - (\tau \cup \sigma)$$



COLLAPSING A VECTOR FIELD

Given a vector field we can simplicially collapse following the arrows.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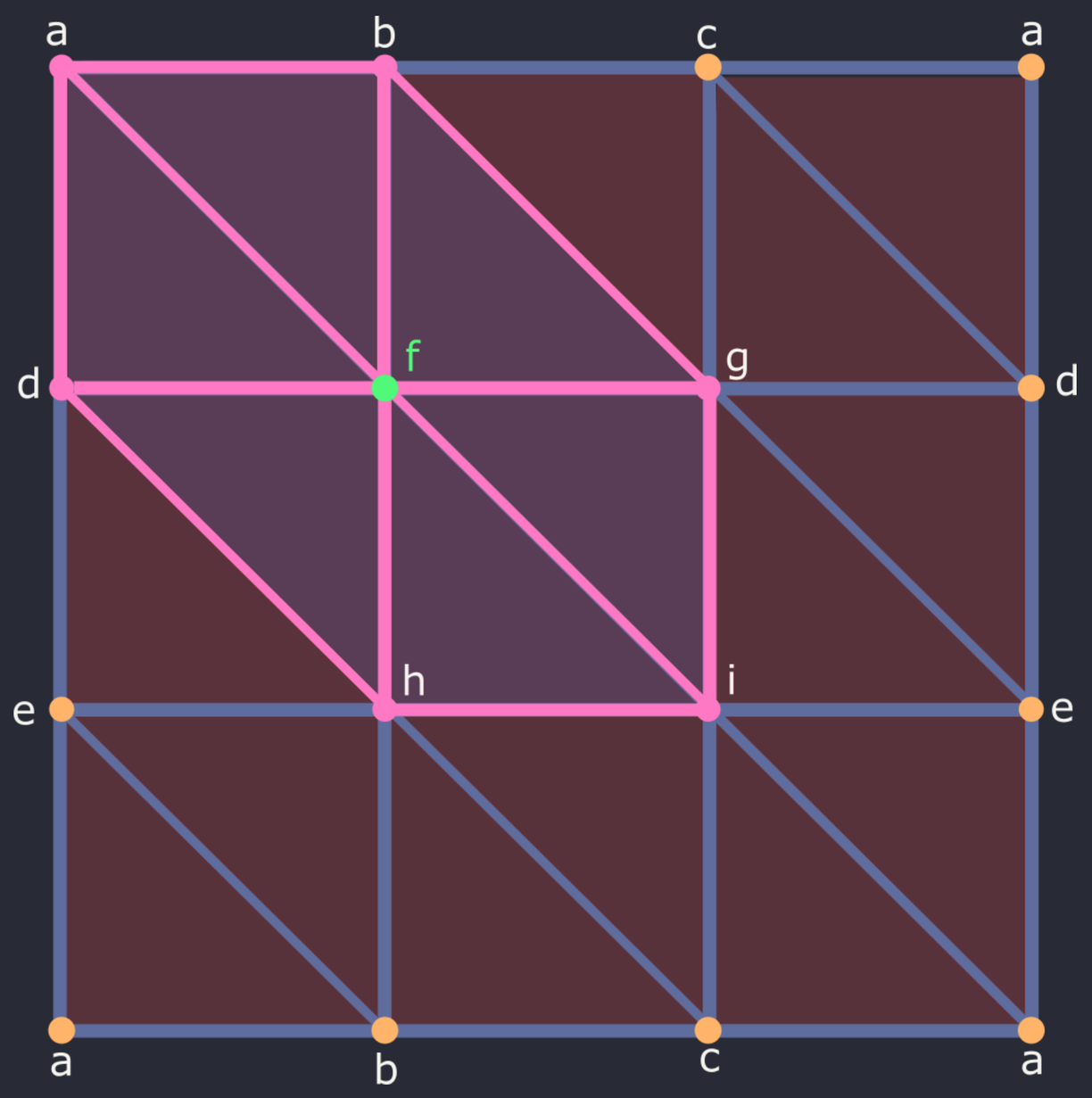
MAIN RESULT OF DISCRETE MORSE THEORY

Suppose K is a simplicial complex with a discrete Morse function. Then K is homotopy equivalent to a CW complex with exactly one cell of dimension n for each critical simplex of dimension n.

SPHERE THEOREMS

STAR OF A FACE

The star of a face τ of K is the subcomplex of K consisting of all faces σ of K where $\tau < \sigma$, as well as all faces of σ.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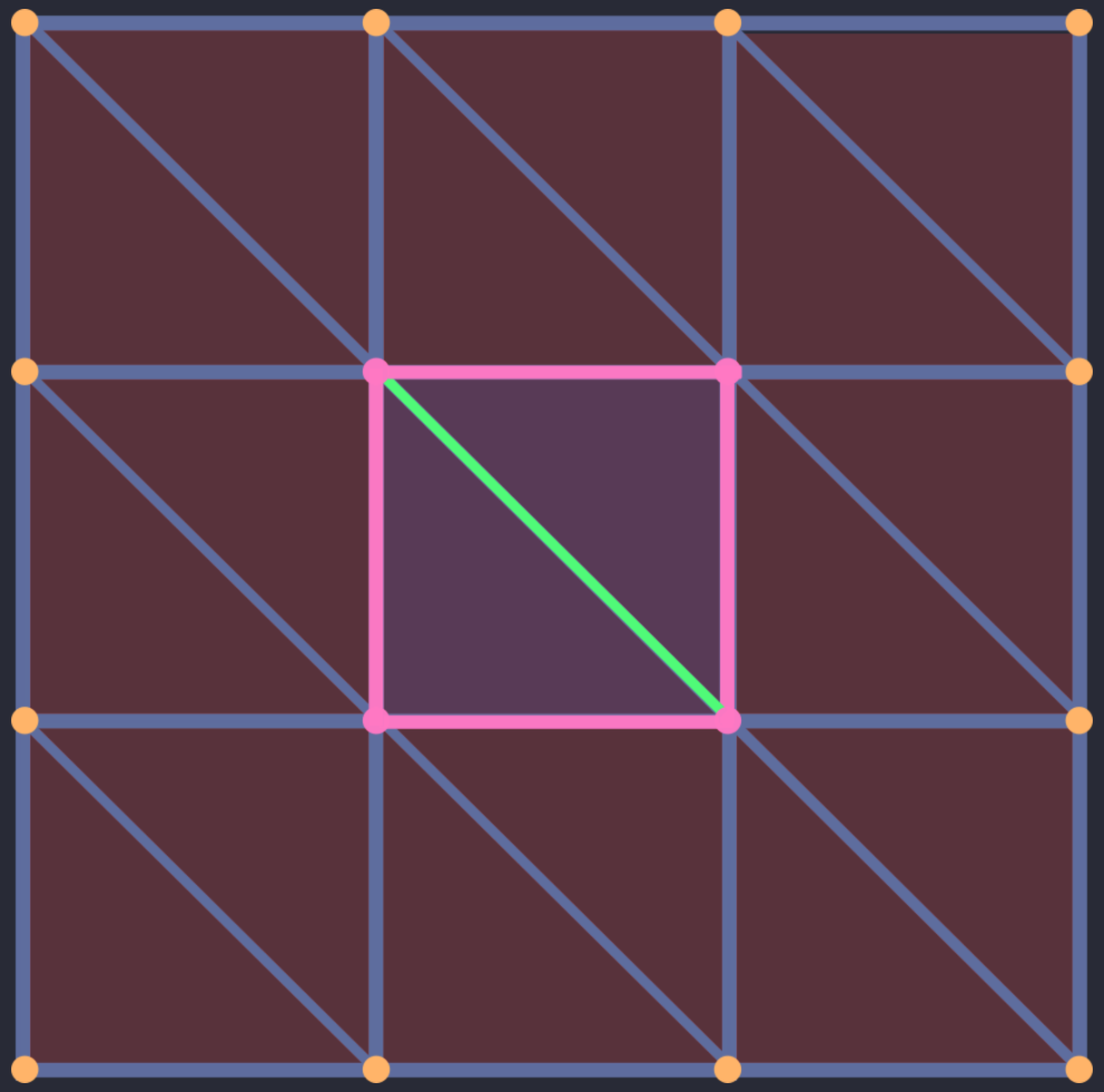
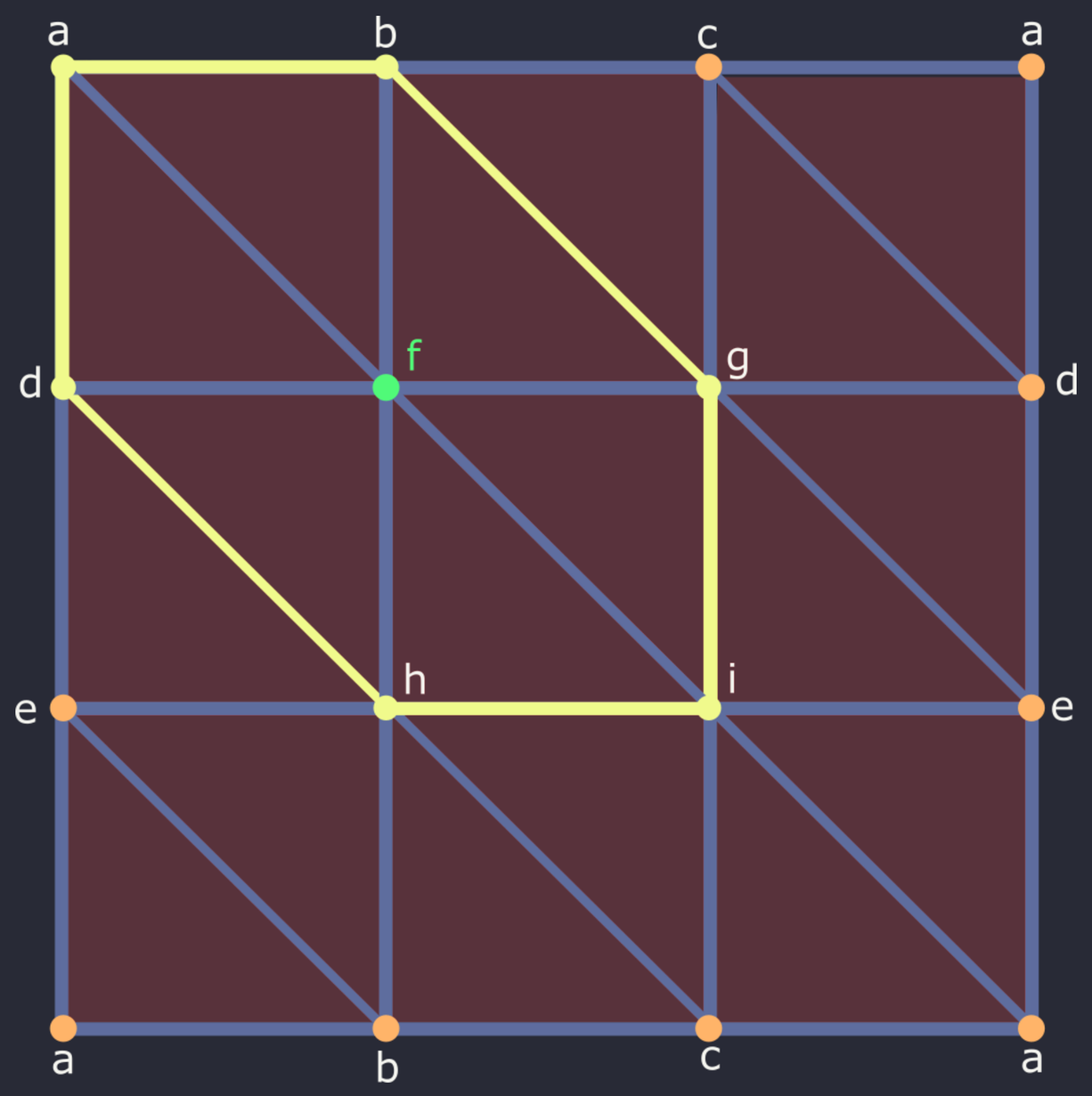


LINK OF A FACE

The link of a face τ of K is the subcomplex of K consisting of all faces of the star of τ that do not intersect τ.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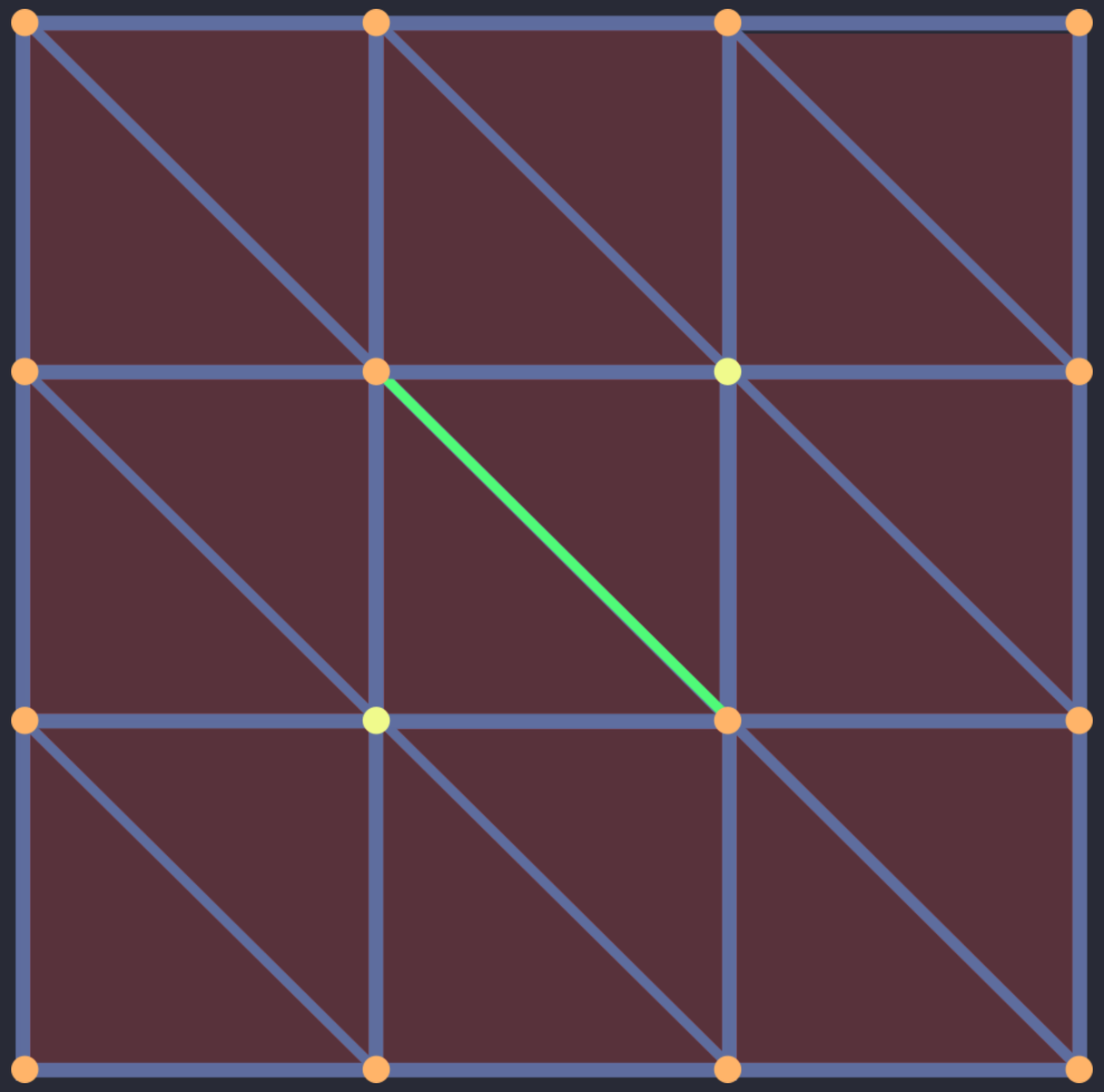

COMBINATORIAL D-BALL

A complex K is a combinatorial d-ball if K and Δ^d have isomorphic subdivisions.

A combinatorial $(d-1)$-sphere is the boundary of a d-ball

COMBINATORIAL D-MANIFOLD

A complex is a d-manifold if the link of every vertex is either a $(d-1)$-ball or $(d-1)$-sphere.

SPHERE THEOREMS

Let K be a combinitorial d-manifold with boundary which simplicially collapses to a vertex. Then K is a combinitorial d-ball.

Let X be a combinatorial d-manifold with a discrete with a discrete Morse function with exactly two critical simplices. Then X is a combinitorial sphere.

REFERENCES

- Robin Forman, A user's guide to discrete Morse theory
- Robin Forman, Morse Theory for Cell Complexes
- Jean Gallier & Jocelyn Quaintance, Aspects of Convex Geometry